```
disp('Sampling theorem and observe aliasing effect')
```

```
Sampling theorem and observe aliasing effect
```

```
Fm=input('Enter the modulating frequency Fm= ');
A=input ('Enter the amplitude A= ')
```

```
A = 50
```

```
t=0:0.1:100;
X=A*sin(2*3.14*Fm*t)
```

```
X = 1×1001
         0   -1.5924   -3.1832   -4.7707   -6.3534   -7.9296   -9.4979  -11.0564 ···
```

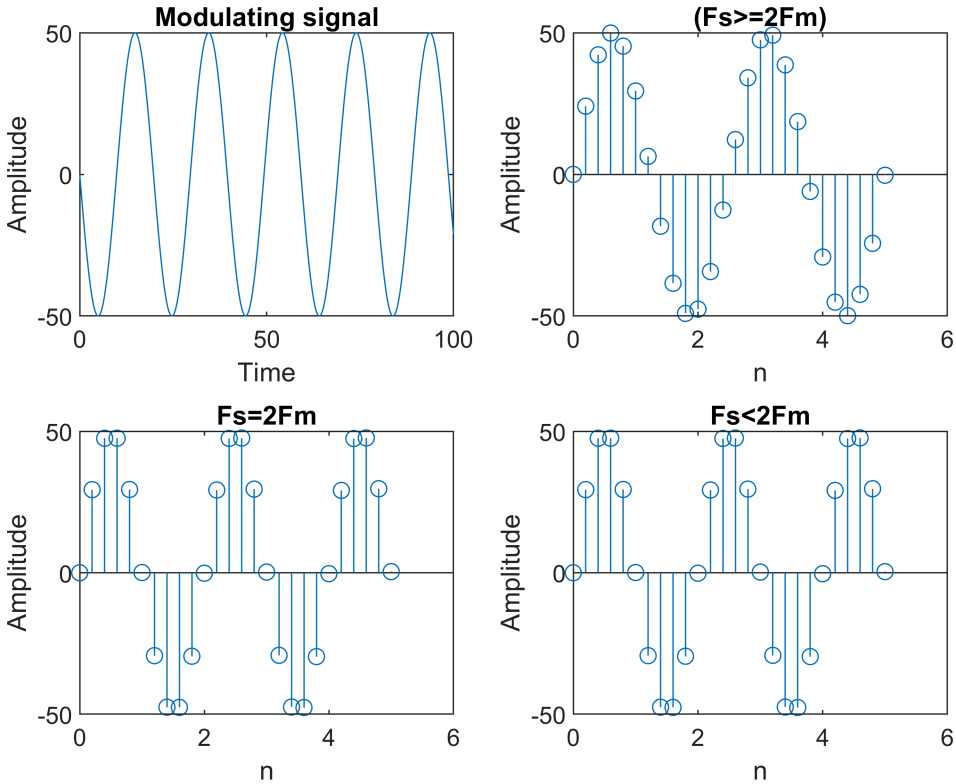
```
subplot(2,2,1);
plot(t,X);
title('Modulating signal');
xlabel('Time')
ylabel('Amplitude')
Fs1=input('Enter the sampling frequency Fs1 (Fs>2fm)= ');
n=0:0.2:5;
y1=A*sin(2*3.14*Fm*n/Fs1)
```

```
y1 = 1×26
         0   24.0765   42.2027   49.8989   45.2630   29.4408    6.3425  -18.3233 ···
```

```
subplot(2,2,2);
stem(n,y1);
title('(Fs>=2Fm)');
xlabel('n')
ylabel('Amplitude')
Fs2=input("Enter the sampling frequency {Fs2(Fs=2Fm)}= ")
```

```
Fs2 = 200
```

```
n=0:0.2:5;
y2=A*sin(2*3.14*Fm*(n/Fs2));
subplot(2,2,3);
stem(n,y2);
title('Fs=2Fm');
xlabel('n');
ylabel('Amplitude');
Fs3=input('Enter the sampling frequency {Fs3(Fs=2Fm)}= ');
n=0:0.2:5;
y3=A*sin(2*3.14*Fm*n/Fs3);
subplot(2,2,4);
stem(n,y2);
title('Fs<2Fm');
xlabel('n');
ylabel('Amplitude');
```


Modulating signal
Amplitude
Time
(Fs>=2Fm)
Amplitude
n
Fs=2Fm
Amplitude
n
Fs<2Fm
Amplitude
n